English

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ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2024-2025)

Telegram: @NEETxNOAH

Test Pattern: NEET (UG)

MAJOR

03-12-2024

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE -MEA, B, C, D, L, M, N, O, P & Q

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to **fill 0012345678.**

Test Booklet Code

E3

This Booklet contains **28** pages.

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology). 50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
   (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
   (b) **Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.
   Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
3. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720**.
4. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
8. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
9. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
10. No candidate, without special permission of the Invigilator, would leave his/her seat.
11. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
12. Use of Electronic/Manual Calculator is prohibited.
13. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
14. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
15. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
16. Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ **Invigilator's Signature :** _____

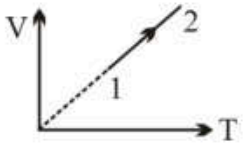
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SECTION-A (PHYSICS)

1. An ideal gas undergoes the process 1 ⟶ 2 as shown in the figure, the heat supplied and work done in the process is ΔQ and ΔW respectively. The ratio $\Delta Q : \Delta W$ is



(1) $\gamma : \gamma - 1$ (2) γ

(3) $\gamma - 1$ (4) $\gamma - 1 : \gamma$

2. ***Assertion :*** When a bottle of cold carbonated drink is opened, a slight fog forms around the opening.
Reason : Adiabatic expansion of the gas causes lowering of temperature which start condensation of water vapours.

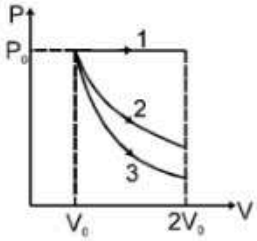
(1) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(2) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

3. A gas is expanded from volume V_0 to $2V_0$ under three different processes. Process 1 is isobaric process, process 2 is isothermal and process 3 is adiabatic. Let ΔU_1, ΔU_2 and ΔU_3 be the change in internal energy of the gas is these three processes. Then :



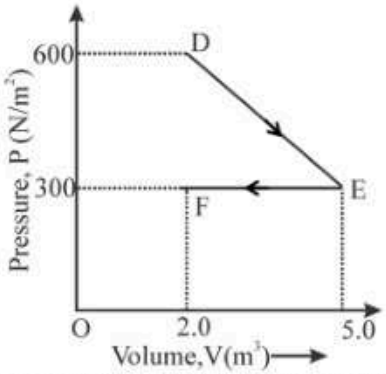
(1) $\Delta U_1 > \Delta U_2 > \Delta U_3$

(2) $\Delta U_1 < \Delta U_2 < \Delta U_3$

(3) $\Delta U_2 < \Delta U_1 < \Delta U_3$

(4) $\Delta U_2 < \Delta U_3 < \Delta U_1$

4. A thermodynamic system is taken from an original state (D) to an intermediate state (E) by the linear process shown in figure:



Its volume is then reduced to the original value from E to F by an isobaric process. Calculate the total work done by the gas from D to E and E to F

(1) 450 J (2) 1350 J

(3) – 1350 J (4) 900 J

5. Molar specific heat of water is ?

(1) 2R (2) 3R (3) 9R (4) R

6. A carnot's engine works between a source at a temperature of 27°C, and a sink at −123°C. Its efficiency is :-

(1) 0.5 (2) 0.25

(3) 0.75 (4) 0.4

7. Oxygen and hydrogen are at the same temperature *T*. The ratio of the mean kinetic energy of oxygen molecules to that of the hydrogen molecules will be:

(1) 16 : 1 (2) 1 : 1

(3) 4 : 1 (4) 1 : 4

8. The temperature of an ideal gas is increased from 27°C to 927°C. The rms speed of its molecules becomes:

(1) Twice

(2) Half

(3) Four times

(4) One-fourth

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9. A gas at 300 K is enclosed in container. Now container is placed on fast moving train while train is in motion, the temperature of gas :

(1) rise above 300 K (2) fall below 300 K

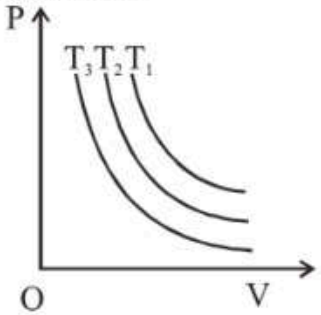
(3) remain unchanged (4) become unsteady

10. If the pressure of an ideal gas contained in a closed vessel is increased by 0.5%, the increase in temperature is 2 K. The initial temperature of the gas is :

(1) 27°C (2) 127°C

(3) 300°C (4) 400°C

11. The isothermal diagram of a gas at three different temperature T_1, T_2 and T_3 is shown in the given figure, Then :-



(1) $T_1 < T_2 < T_3$ (2) $T_1 < T_2 > T_3$

(3) $T_1 > T_2 > T_3$ (4) $T_1 > T_2 < T_3$

12. 2 mole of an ideal monoatomic gas at temperature T_0 are mixed with 4 mole of another ideal monoatomic gas at temperature 2 T_0 , then the temperature of mixture is

(1) $\frac{5}{3}T_0$ (2) $\frac{3}{2}T_0$

(3) $\frac{4}{3}T_0$ (4) $\frac{5}{4}T_0$

13. For O_2 gas, $C_p - C_v = a$ and for N_2 gas; $C_p - C_v = b$, if C_p & C_v are gram specific heat then ratio of a to b (a : b) is -

(1) $\frac{7}{8}$ (2) $\frac{8}{7}$

(3) $\frac{1}{1}$ (4) None of these

14. During an adiabatic process the pressure of the gas is found to be proportional to the cube of the absolute temperature. The ratio $C_P/C_V = \gamma$ for the gas is:

(1) 3/2 (2) 2

(3) 4/3 (4) 5/3

15. If a metallic sphere gets cooled from 62°C to 50°C in 10 minutes and in the next 10 minutes gets cooled to 42°C, then the temperature of the surroundings is :

(1) 30°C (2) 36°C

(3) 26°C (4) 20°C

16. A black body has maximum wavelength λ_m at 2000 K. Its corresponding wavelength at 3000 K will be:

(1) $\frac{16}{81}\lambda_m$ (2) $\frac{81}{16}\lambda_m$

(3) $\frac{3}{2}\lambda_m$ (4) $\frac{2}{3}\lambda_m$

17. A spring elongated by length 'ℓ' when a mass 'M' is suspended to it. Now a tiny mass 'm' is attached and then released, its time period of oscillation is:-

(1) $2\pi\sqrt{\frac{(M+m)\ell}{Mg}}$

(2) $2\pi\sqrt{\frac{m\ell}{Mg}}$

(3) $2\pi\sqrt{\ell/g}$

(4) $2\pi\sqrt{\frac{M\ell}{(m+M)g}}$

18. The time period of a mass suspended from a spring is 5 sec. The spring is cut into four equal parts and the same mass is now suspended from one of its parts. The period is now :-

(1) 5 s (2) 2.5 s

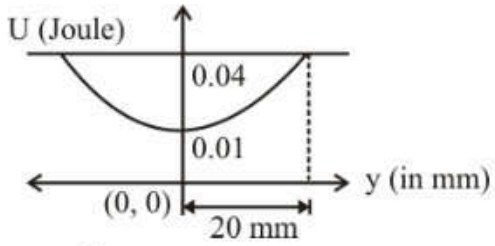
(3) 1.25 s (4) (5/16) s

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19. A simple pendulum performs simple harmonic motion about x = 0 with an amplitude a and time period T. The speed of the pendulum at $x = \frac{a}{\sqrt{2}}$ will be :-

(1) $\frac{3\pi a}{T}$ (2) $\frac{4\pi a}{T}$ (3) $\frac{\sqrt{2}\pi a}{T}$ (4) $\frac{aT}{\pi^2}$

20. The variation of potential energy of harmonic oscillator is as shown in figure. The spring constant is



(1) 1×10^2 N/m

(2) 150 N/m

(3) 0.667×10^2 N/m

(4) 3×10^2 N/m

21. Two waves are represented by : $y_1 = 4 \sin 404\ \pi t$ and $y_2 = 3 \sin 400\ \pi t$. Then :

(1) beat frequency is 4 Hz and the ratio of maximum to minimum intensity if 49 : 1

(2) beat frequency is 2 Hz and the ratio of maximum to minimum intensity is 49 : 1

(3) beat frequency is 2 Hz and the ratio of maximum to minimum intensity is 1 : 49

(4) beat frequency is 4 Hz and the ratio of maximum to minimum intensity is 1 : 49

22. A sonometer wire supports 4 kg load and vibrates in fundamental mode with tunning fork of frequency 416 Hz. The length of wire between the bridges is now doubled. In order to maintain fundamental mode the load should be :-

(1) 1 kg (2) 2 kg

(3) 8 kg (4) 16 kg

23. When a wave travels in a medium, the particle displacement is given by the equation $y = a \sin 2\pi(bt - cx)$ where a, b and c are constants. The maximum particle velocity will be twice the wave velocity if :-

(1) $c = \frac{1}{\pi a}$ (2) $c = \pi a$

(3) $b = ac$ (4) $b = \frac{1}{ac}$

24. A string of length 3 m and linear mass density 0.0025 kg/m is fixed at both ends. One of its resonance frequency is 252 Hz. The next higher resonance frequency is 336 Hz. Then the fundamental frequency will be -

(1) 84 Hz (2) 63 Hz

(3) 126 Hz (4) 168 Hz

25. In a resonance tube experiment, the first resonance is obtained for 10 cm of air column and the second for 32 cm. The end correction for this apparatus is–

(1) 0.5 cm (2) 1.0 cm

(3) 1.5 cm (4) 2 cm

26. A student is performing experiment with resonance tube apparatus in Physics lab to find the speed of sound at room temperature. He got first two resonating lengths of air column as 17 cm and 51 cm, using tuning fork of frequency 512 Hz. Find speed of sound at room temperature.

(1) 348 m/s

(2) 358 m/s

(3) 332 m/s

(4) 322 m/s

27. 100 g of ice at 0°C is mixed with 100 g of water at 100°C. What will be the final temperature of the mixture ?

(1) 10°C (2) 20°C

(3) 30°C (4) 40°C

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28. Steam at 100°C is added slowly to 1400 gm of water at 16°C until the temperature of water is raised to 80°C. The mass of steam required :-

(1) 120 gm (2) 240 gm

(3) 160 gm (4) 320 gm

29. The ratio of the speed of sound in nitrogen gas to that in helium gas at 300 K is :

(1) $\sqrt{\frac{2}{7}}$ (2) $\sqrt{\frac{1}{7}}$

(3) $\frac{\sqrt{3}}{5}$ (4) $\frac{\sqrt{6}}{5}$

30. The period of a simple pendulum inside a stationary lift is T. The lift accelerates upwards with an acceleration of $\frac{g}{3}$.The time period of pendulum will be :-

(1) $\sqrt{2}T$ (2) $\frac{T}{\sqrt{2}}$ (3) $\frac{\sqrt{3}}{2}T$ (4) $\frac{T}{3}$

31. The mass and diameter of a planet are twice those of earth. The period of oscillation of second pendulum on this planet will be :

(1) $\frac{1}{\sqrt{2}}$s (2) $2\sqrt{2}$s (3) 2s (4) $\frac{1}{2}$s

32. A pendulum is hung from the roof of a sufficiently high building and is moving freely to and fro like a simple harmonic oscillator. The acceleration of the bob of the pendulum is 20 m/s^2 at a distance of 5 m from the mean position. The time period of oscillation is :-

(1) 2π s (2) π s

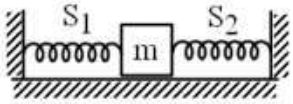
(3) 2 s (4) 1 s

33. A particle executes simple harmonic motion along a straight line with an amplitude A. The potential energy is maximum when the displacement is :

(1) $\pm$ A (2) Zero

(3) $\pm\frac{A}{2}$ (4) $\pm\frac{A}{\sqrt{2}}$

34. When a mass m is connected individually to two springs S_1 and S_2, the oscillation frequencies are V_1 and V_2, if the same mass is attached to the two springs in given figure, the oscillation frequency would be :



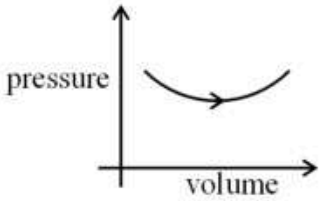
(1) $V_1 + V_2$

(2) $\sqrt{V_1^2 + V_2^2}$

(3) $\left(\frac{1}{V_1} + \frac{1}{V_2}\right)^{-1}$

(4) $\sqrt{V_1^2 - V_2^2}$

35. Consider the process on a system shown in figure below. During the process, the rate with which the work is being done by the system :



(1) Continuously increases

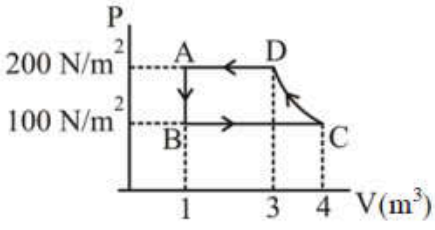
(2) Continuously decreases

(3) First increases then decreases

(4) First decreases then increases

SECTION-B (PHYSICS)

36. The P-V diagram of a diatomic ideal gas system going under cyclic process is as shown. The work done during an adiabatic process CD is (γ = 1.4)



(1) −500 J (2) −400 J

(3) 400 J (4) 200 J

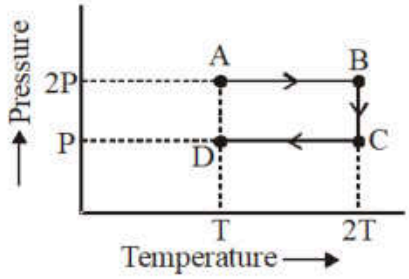
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37. A thermally insulated vessel contain an ideal gas of molar mass M and ratio of specific heats γ. It is moving with speed v and is suddenly brought to rest. Assuming no heat is lost to the surroundings. Its temperature increased by :-

(1) $\frac{\gamma-1}{2(\gamma-1)R}MV^2$ (2) $\frac{(\gamma-1)}{2\gamma R}MV^2$

(3) $\frac{\gamma MV^2}{2R}$ (4) $\frac{(\gamma-1)}{2R}MV^2$

38. An ideal monoatomic gas is taken through the thermodynamic states A⟶B⟶C⟶D via the paths shown in the figure. If U_A,U_B,U_C and U_D represent the internal energy of the gas in state A,B,C and D respectively, then which of the following is not true?



(1) $U_A = U_D$

(2) $U_B = U_A$

(3) $U_B = U_C$

(4) $U_C \neq U_D$

39. A container of volume $1m^3$ is divided into two equal compartments by a partition. One of these compartments contains an ideal gas at 300 *K*. The other compartment is vacuum. The whole system is thermally isolated from its surroundings. The partition is removed and the gas expands to occupy the whole volume of the container. Its temperature now would be :

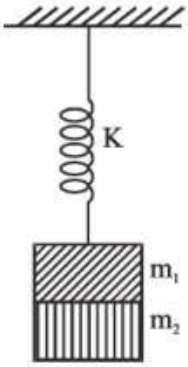
(1) 300 K

(2) 239 K

(3) 200 K

(4) 100 K

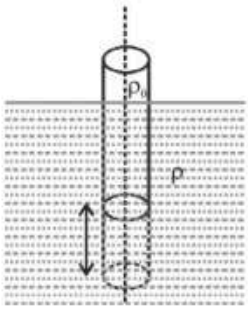
40. Two masses m_1 and m_2 are suspended together by a massless spring of constant k. When the masses are in equilibrium, m_1 is removed without disturbing the system; the amplitude of vibration is:



(1) m_1g/k (2) m_2g/k

(3) $\frac{(m_1+m_2)g}{k}$ (4) $\frac{(m_2-m_1)g}{k}$

41. A solid cylinder of density ρ_0, cross-section area A and length ℓ floats in a liquid $\rho(>\rho_0)$ with its axis vertical, as shown. If it is slightly displaced downward and released, the time period will be :-



(1) $2\pi\sqrt{\frac{\ell}{g}}$ (2) $2\pi\sqrt{\frac{\rho_0\ell}{\rho g}}$

(3) $2\pi\sqrt{\frac{\rho\ell}{\rho_0 g}}$ (4) $2\pi\sqrt{\frac{\ell}{2g}}$

42. The amplitude of a particle executing SHM is 4 cm. At the mean position the speed of the particle is 16 cm/sec. The distance of the particle from the mean position at which the speed of the particle becomes $8\sqrt{3}$cm/s, will be :-

(1) $2\sqrt{3}$cm (2) $\sqrt{3}$ cm

(3) 1cm (4) 2cm

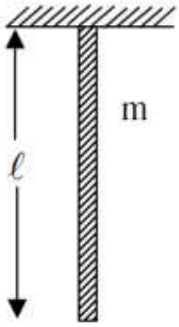
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43. A simple pendulum has a time period T in vacuum. Its time period when it is completely immersed in a liquid of density one-eight of the density of material of the bob is :-

(1) $\sqrt{\frac{7}{8}}T$ (2) $\sqrt{\frac{5}{8}}T$

(3) $\sqrt{\frac{3}{8}}T$ (4) $\sqrt{\frac{8}{7}}T$

44. A uniform rope of mass m and length ℓ hangs from a ceiling as shown in figure. Find out time taken by a transverse wave to travel the full length of rope.



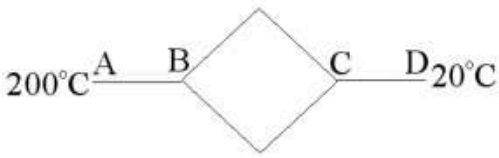
(1) $t = \sqrt{\frac{\ell}{g}}$ (2) $t = \sqrt{\frac{g}{\ell}}$

(3) $t = 2\sqrt{\frac{\ell}{g}}$ (4) $t = \frac{1}{2}\sqrt{\frac{\ell}{g}}$

45. An open organ pipe of length ℓ is sounded together with another open organ pipe of length $\ell + x$ in their fundamental tones. Speed of sound in air is v. The beat frequency heard will be : ($x \ll \ell$) :-

(1) $\frac{vx}{4\ell^2}$ (2) $\frac{v\ell^2}{2x}$ (3) $\frac{vx}{2\ell^2}$ (4) $\frac{vx^2}{2\ell}$

46. Six identical conducting rods are joined as shown in figure. Points A and D are maintained at temperature 200°C and 20°C respectively. The temperature of the junction B will be :-



(1) 120°C (2) 100°C

(3) 140°C (4) 80°C

47. An organ pipe filled with a gas at 27°C resonates at 400 Hz in its fundamental mode. If it is filled with the same gas at 90°C, the resonance frequency at the same mode will be :-

(1) 420 Hz (2) 440 Hz

(3) 484 Hz (4) 512 Hz

48. A source of frequency 2400 Hz when vibrated over the mouth of a closed organ pipe is in unison at 300 K. The beats produced per second when temperature rises by 1 K is :

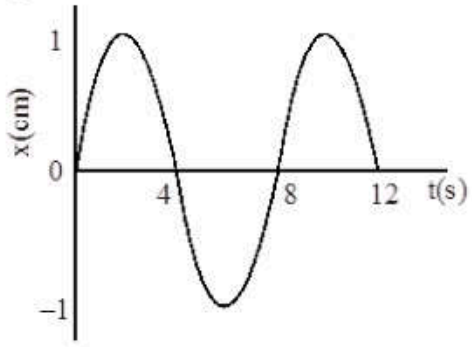
(1) 8 (2) 6 (3) 4 (4) 2

49. If two tuning forks A and B are sounded together, they produce 4 beats per second. A is then slightly loaded with wax, they produce same beats when sounded again. The initial frequency of A is 256 Hz. The frequency of B will be :-

(1) 250 Hz (2) 252 Hz

(3) 260 Hz (4) 262 Hz

50. The x-t graph of a particle undergoing simple harmonic motion is shown below. The acceleration of the particle at $t = 4/3$ s is



(1) $\frac{\sqrt{3}}{32}\pi^2$ cm/s^2

(2) $\frac{-\pi^2}{32}$ cm/s^2

(3) $\frac{\pi^2}{32}$ cm/s^2

(4) $-\frac{\sqrt{3}}{32}\pi^2$ cm/s^2

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SECTION-A (CHEMISTRY)

51. Which of the following homologous series has incorrect general formula :-

(1) Alkyne C_nH_{2n-2}

(2) Alkanol $C_nH_{2n+2}O$

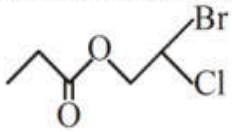
(3) Alkanal $C_nH_{2n+1}O$

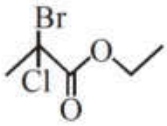
(4) Carboxylic acid $C_nH_{2n}O_2$

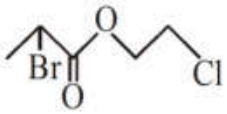
52. Minimum number of carbon atoms present in an ester are :-

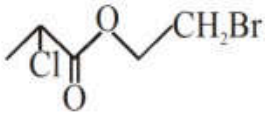
(1) 2 (2) 1 (3) 4 (4) 3

53. Structure of the compound
Ethyl 2-bromo-2-chloropropanoate is :-

(1) 

(2) 

(3) 

(4) 

54.

Mixture		Purification technique	
(A)	Chloroform and Aniline	(P)	Distillation
(B)	Glycerol and spentlye	(Q)	Chromatography
(C)	Aniline and water	(R)	Steam Distillation
(D)	Amino acid	(S)	Distillation under Reduced pressure

Correct Match:-

(1) A-P, B-Q, C-R, D-S

(2) A-P, B-S, C-R, D-Q

(3) A-P, B-R, C-S, D-Q

(4) A-R, B-S, C-P, D-Q

55. (I) $CH_3-CH_2-NH-CHO$ (II) $CH_3-\underset{\underset{NH_2}{|}}{CH}-CHO$

Which type of isomerism is observed between I & II :

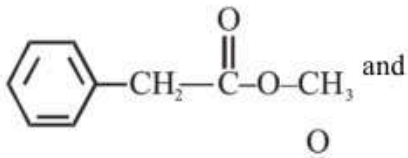
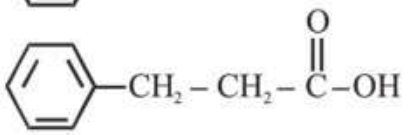
(1) Chain isomerism

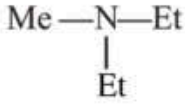
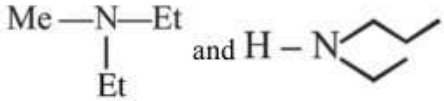
(2) Position isomerism

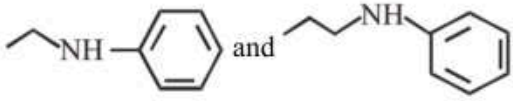
(3) Functional isomerism

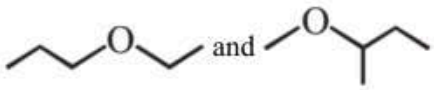
(4) Metamerism

56. Which of the following is a pair of metamers.

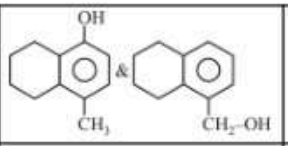
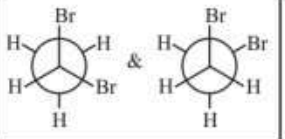
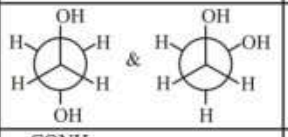
(1)  and 

(2)  and 

(3) and 

(4) and 

57. Match the column :

(A)		(P)	Position isomer
(B)		(Q)	Functional group isomer
(C)		(R)	Chain isomer
(D)		(S)	Conformer

(1) A-P,B-Q,C-R,D-S

(2) A-Q, B-P, C-R, D-S

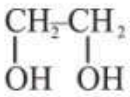
(3) A-Q,B-P,C-S,D-R

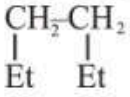
(4) A-S,B-P,C-S,D-P

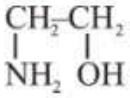
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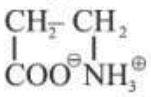
Telegram: @NEETxNOAH

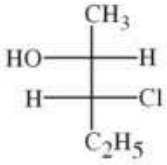
58. In which of the following gauche form will be not most stable form :-

(1) 

(2) 

(3) 

(4) 

59. The full name of compound  is :-

(1) (2R, 3R)- 3-chloro-2-pentanol

(2) (2R, 3S)- 3-chloro-2-pentanol

(3) (2S, 3R)- 3-chloro-2-pentanol

(4) (2S, 3S)- 3-chloro-2-pentanol

60. Which one of the following series contains electrophiles only ?

(1) H_2O, SO_3, H_3O^+

(2) NH_3, H_2O, $AlCl_3$

(3) $AlCl_3$, SO_3, $\overset{+}{N}O_2$

(4) H_2O, $\overset{+}{Cl}$, NH_3

61. Which of the following is the correct order for –I effect ?

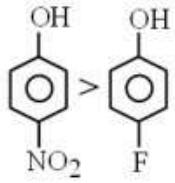
(1) $-NO_2 > -CN > -F > -C_6H_5$

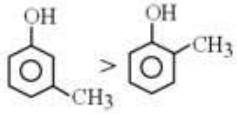
(2) $-C_6H_5 > -F > -CN > -NO_2$

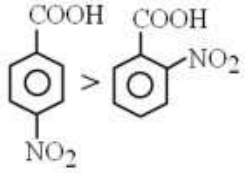
(3) $-F > -CN > -C_6H_5 > -NO_2$

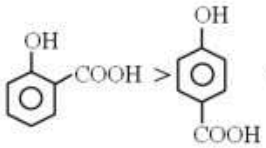
(4) $-CN > C_6H_5 > -NO_2 > -F$

62. Which of the following show incorrect relation?

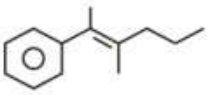
(1)  ⇒ Acidic strength order

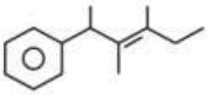
(2)  ⇒ Acidic strength order

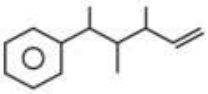
(3)  ⇒ Acidic strength order

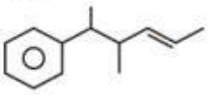
(4)  ⇒ Acidic strength order

63. Which of the following has least heat of hydrogenation ?

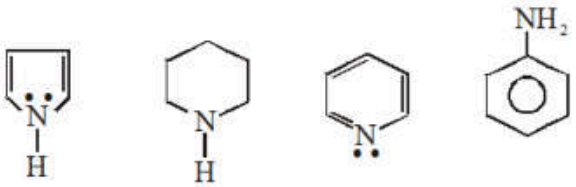
(1) 

(2) 

(3) 

(4) 

64. Correct decreasing order of basic strength for following:-



(I) (II) (III) (IV)

(1) I > II > III > IV

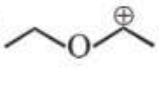
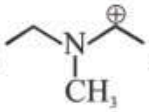
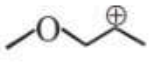
(2) II > III > I > IV

(3) II > IV > I > III

(4) II > III > IV > I

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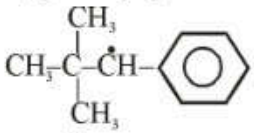
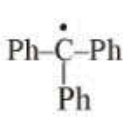
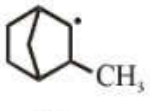
65. (a)  (b)  (c) 

Which is correct order for stability of above carbocations :-

(1) c > a > b (2) b > a > c

(3) c > b > a (4) b > c > a

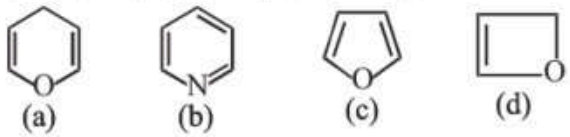
66. Hyperconjugation occurs in :-

 (i)  (ii)  (iii)

(1) Only (i) (2) (i) & (ii)

(3) (i), (ii) & (iii) (4) only (iii)

67. Which of the following is aromatic :-

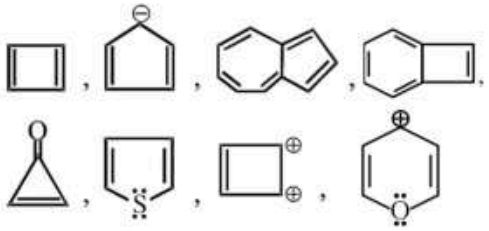


(a) (b) (c) (d)

(1) a and b (2) b and c

(3) c and d (4) a and d

68. Number of aromatic compounds in given compounds :-

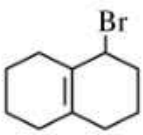


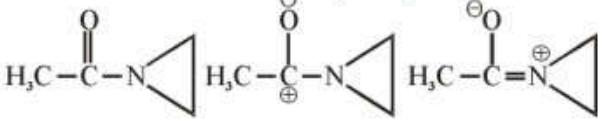
(1) 6 (2) 8

(3) 4 (4) 5

69. which of the following is maximum reactive towards De-hydrohalogination

(1)  (2) 

(3)  (4) 

70. 

(X) (Y) (Z)

The correct stability order of the given canonical structure is :

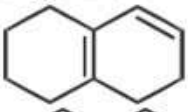
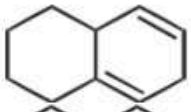
(1) X > Y > Z

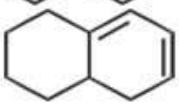
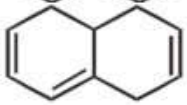
(2) Z > X > Y

(3) X > Z > Y

(4) Y > Z > X

71. Rank the following in decreasing order of heat of hydrogenation :-

(i)  (ii) 

(iii)  (iv) 

(1) iii > i > ii > iv

(2) i > iii > ii > iv

(3) iv > ii > iii > i

(4) iv > iii > ii > i

72. (I)  (II)  (III) 

Correct statement about these three compounds :-

(1) Stability order : II > III > I

(2) Stability order : III > I > II

(3) Heat of combustion : I > III > II

(4) Both (1) and (3) statements are correct

73. The correct order of boiling point of following compounds is :

A) n-pentane B) Isopentane C) Neopentane

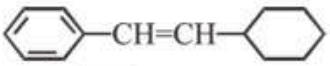
(1) A > B > C

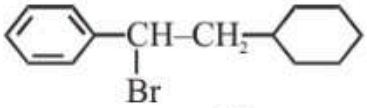
(2) B > A > C

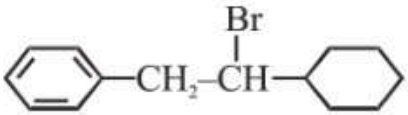
(3) C > B > A

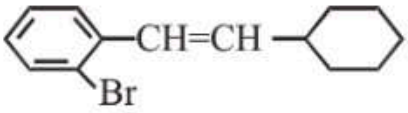
(4) A > C > B

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74.  ? (Major product)

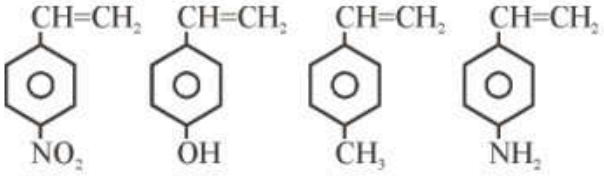
(1) 

(2) 

(3) 

(4) Both (1) & (2)

75. Decreasing order of rate of reaction of molecules towards Electrophilic addition reaction is :-



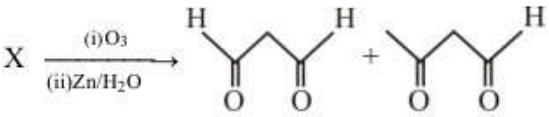
(a) (b) (c) (d)

(1) b > c > a > d

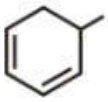
(2) a > b > c > d

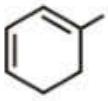
(3) d > b > c > a

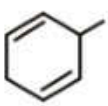
(4) b > d > c > a

76. 

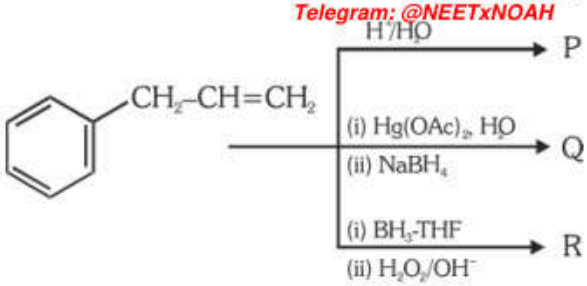
Indentify X in above reaction .:-

(1) 

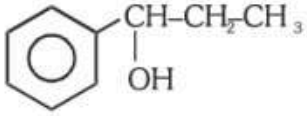
(2) 

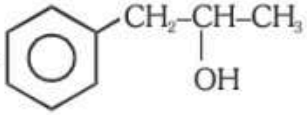
(3) 

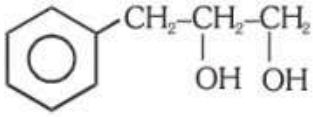
(4) 

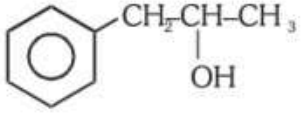
77. 

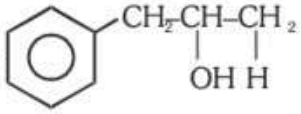
Products P, Q and R are respectively –

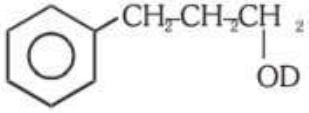
(1) P = 

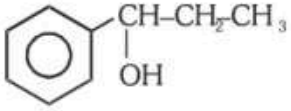
Q = 

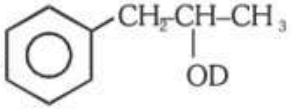
R = 

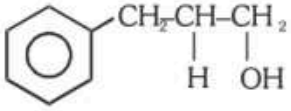
(2) P = 

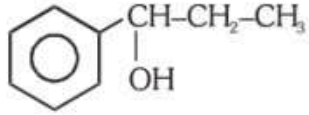
Q = 

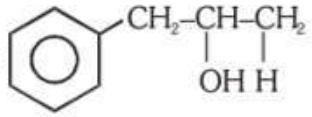
R = 

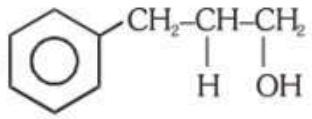
(3) P = 

Q = 

R = 

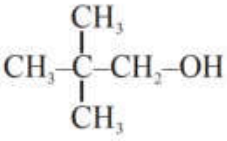
(4) P = 

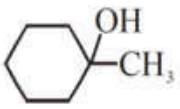
Q = 

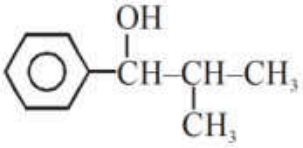
R = 

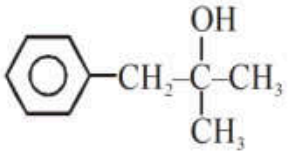
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78. Which of the following alcohol undergoes dehydration reaction most easily :

(1) 

(2) 

(3) 

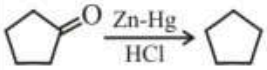
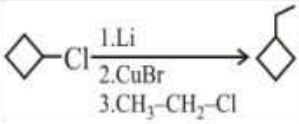
(4) 

79. Reactivity of halogen in halogenation:-

(1) $F_2 < Cl_2 < Br_2 < I_2$ (2) $Cl_2 > F_2 > Br_2 > I_2$

(3) $F_2 > Cl_2 > Br_2 > I_2$ (4) $Br_2 > Cl_2 > F_2 > I_2$

80. Match the following columns -

	Column-I (Reaction)		Column-II (Name of reaction)
A.	$CH_3-CH_2-Br \xrightarrow[\text{Dry ether}]{Na}$ n-Butane	P	Clemmensen's reduction
B.	$\begin{matrix}CH_2-COONa\\ \vert\\ CH_2-COONa\end{matrix} \xrightarrow[\text{Electrolysis}]{NaOH}$ Ethene	Q	Correy House synthesis
C.		R	Kolbe's electrolysis
D.		S	Wurtz reaction

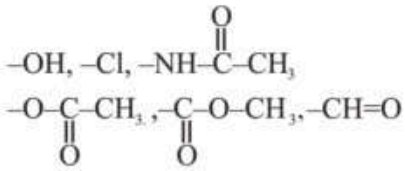
(1) A-S, B-R, C-Q, D-P

(2) A-R, B-S, C-P, D-Q

(3) A-Q, B-R, C-P, D-S

(4) A-S, B-R, C-P, D-Q

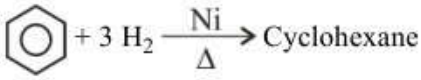
81. How many of following groups are ortho-para directing?

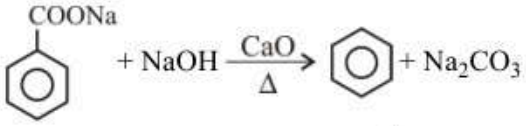


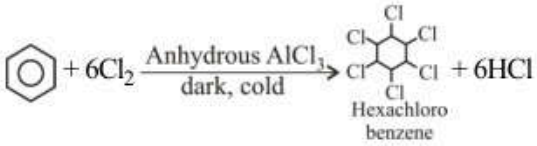
(1) 4 (2) 6 (3) 5 (4) 3

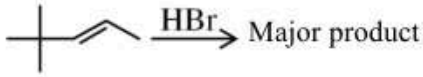
82. Which reaction is incorrect ?

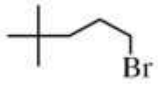
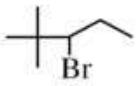
(1) 

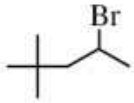
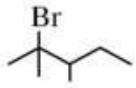
(2) 

(3) 

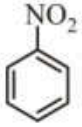
(4) 

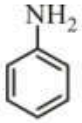
83. 

(1)  (2) 

(3)  (4) 

84. In which of the following compound, Kjeldahl method is applicable :

(1)  (2) 

(3)  (4) Ph–N=N–Ph

85. Mixture of Anthracene and benzoic Acid can be separated by

(1) Fractional distillation

(2) Sublimation

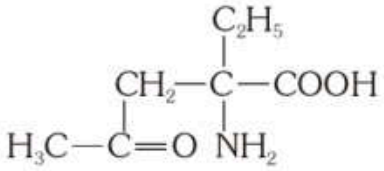
(3) Chromatography

(4) Filteration

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SECTION-B (CHEMISTRY)

86. The IUPAC name of given compound is :-



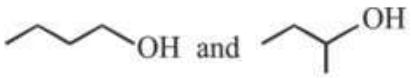
(1) 2–Oxo–4–amino–4–ethyl pentanoic acid

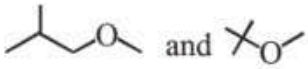
(2) 2–Amino–4–oxo–2–ethyl pentanoic acid

(3) 3–Amino–3–carboxy–5–hexanone

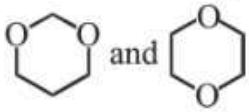
(4) 2–Amino–2–ethyl–4–oxo pentanoic acid

87. Which of the following pairs of compounds are not position isomer :-

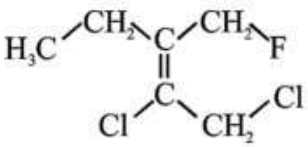
(1) 

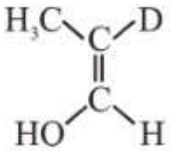
(2) 

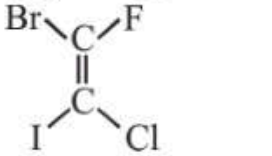
(3) 

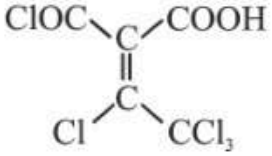
(4) 

88. Which of the following shows E-configuration ?

(1) 

(2) 

(3) 

(4) 

89. **Statement-I** : Molecules having chiral carbons will always be optically active.

Statement-II : it is not necessary to have chiral carbon to show chirality.

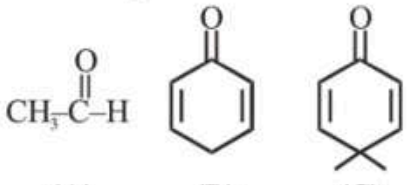
(1) Both statement-I and statement-II are correct

(2) Both statement-I and Statement-II are incorrect

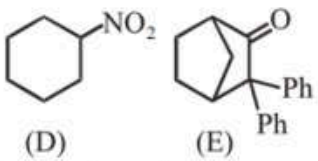
(3) Statement-I is incorrect and statement-II is correct

(4) Statement-I is correct and statement-II is incorrect

90. Consider following molecules :



(A) (B) (C)



(D) (E)

Which can show tautomerism :

(1) A, B, C (2) B, C, E

(3) A, B, D (4) B, C, E, F

91. How many no. of groups showing +M effect is :-

(a) –OH (b) $-\overset{\oplus}{N}H_3$ (c) $-OCH_3$ (d) –F

(1) 4 (2) 3 (3) 2 (4) 1

92. Match the List-I with List-II :-

	List - I (Compounds)		List - II (pK_a- Value)
(a)	o-Nitrophenol	(i)	10
(b)	p-Nitrophenol	(ii)	8.3
(c)	m-Nitrophenol	(iii)	7.2
(d)	Phenol	(iv)	7.1

(1) a-iv, b-iii, c-ii, d-i

(2) a-iv, b-ii, c-iii, d-i

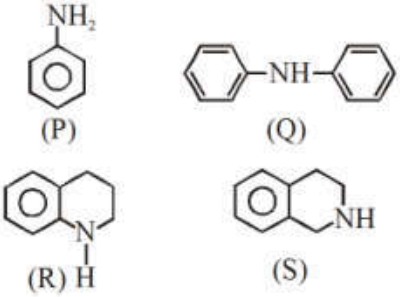
(3) a-iii, b-iv, c-i, d-ii

(4) a-iii, b-iv, c-ii, d-i

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93. Arrange the following according to basicity :-

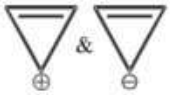


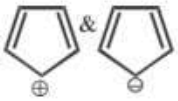
(P) (Q) (R) (S)

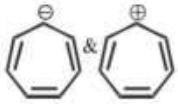
(1) P > S > R > Q (2) S > R > P > Q

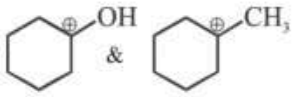
(3) S > Q > R > P (4) S > R > Q > P

94. In which pair second ion is more stable than first?

(i) 

(ii) 

(iii) 

(iv) 

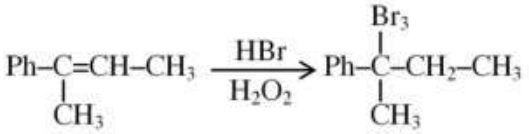
(1) (i) and (ii)

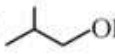
(2) (ii) and (iii)

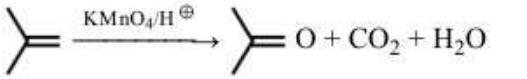
(3) (ii) and (iv)

(4) (iii) and (iv)

95. Which of following reaction is correct :

(1) 

(2) $(CH_3)_3CH \xrightarrow[\text{Oxidation}]{KMnO_4(O)}$  (1°Alc.)

(3) 

(4) $CH_3–C\equiv CH \xrightarrow{Hg^{2+}/dil.H_2SO_4} CH_3–CH_2–CH=O$

96. The correct statement regarding the basicity of arylamines is :-

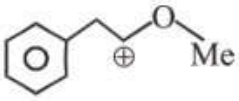
(1) Arylamines are generally less basic than alkylamines because the nitrogen lone-pair electrons are delocalized by interaction with the aromatic ring π electron system.

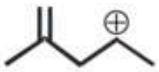
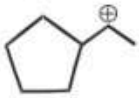
(2) Arylamines are generally more basic than alkylamines because the nitrogen lone-pair electrons are not delocalized by interaction with the aromatic ring π electron system.

(3) Arylamines are generally more basic than alkylamines because of aryl group.

(4) Arylamines are generally more basic than alkylamines, because the nitrongen atom in arylamines is sp-hybridized.

97. Carbocation rearrangement is not observed in

(1)  (2) 

(3)  (4) 

98. Which of the following alcohols on dehydration with conc. H_2SO_4 at 440 K will give propylene ?

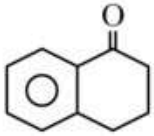
(i) *n*-propyl alcohol (ii) Isobutyl alcohol

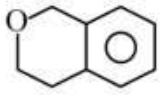
(iii) Isopropyl alcohol (iv) *n*-butyl alcohol

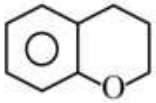
(1) (ii), (iii) (2) (i), (iii)

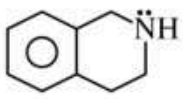
(3) (i), (ii), (iii) (4) (ii), (iii), (iv)

99. Which is maximum reactive towards electrophilic substitution reaction (ESR)

(1) 

(2) 

(3) 

(4) 

100. 0.5 gm of an organic compound gave 50 ml of N_2 collected at 300 K temperature and 715 mm pressure By DUMAS method. Calculate percentage of nitrogen in the compound (Aqueous Tension at 300 K = 15 m.m)

(1) 17.465 (2) 15.899

(3) 12.475 (4) 10.475

SECTION-A (BOTANY)

101. Which of the following statement is incorrect ?

(1) Splitting of water molecule take place on the inner side of membrane of thylakoid.

(2) Primary electron acceptor (Pheophytin) is located towards the outer side of thylakoid membrane.

(3) ATP synthase enzyme consist of two part one called CF_o is embedded in the membrane and form a transmembrane channel.

(4) Cyclic photophosphorylation involves both PSI and PSII.

102. Match the following:

Column-I		Column-II	
(a)	Primary acceptor of CO_2 in C_3 plant	(i)	PEP
(b)	Primary acceptor of CO_2 in C_4 plant	(ii)	PGA
(c)	First CO_2 fixation product in C_3 plant	(iii)	OAA
(d)	First CO_2 fixation product in C_4 plant	(iv)	RUBP

(1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

(2) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

(3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

(4) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

103. **Statement-I :** C_4 plants lack photorespiration.
Statement-II : C_4 plants have a mechanism that increases the concentration of CO_2 at the site of RuBisCO enzyme.

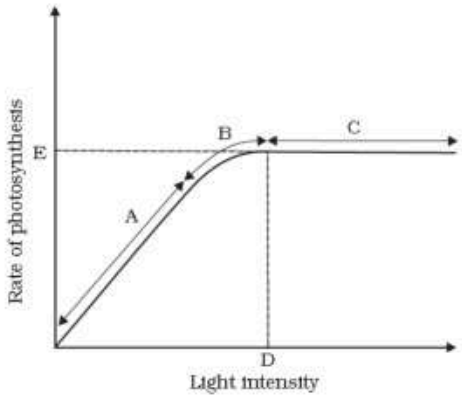
(1) Both **Statement I** and **Statement II** are incorrect.

(2) Both **Statement I** and **Statement II** are correct.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct

104.



What does the "B" label represent :

(1) Light limiting factor

(2) Saturation point of light

(3) Factor other than CO_2 are limiting

(4) Factor other than light are limiting

105. Where exactly enzymes of dark reaction are present within chloroplast ?

(1) Thylakoid membrane

(2) Grana

(3) Stromal lamella

(4) Stroma of chloroplast

106. What is the main factor that limits the rate of photosynthesis on a clear day ?

(1) CO_2 (2) Water

(3) Light (4) Chlorophyll

107. Which of the following organism was used in experiment of T.W. Engelmann ?

(1) *Cladophora* (2) Mint

(3) *Albugo* (4) *Chlorella*

108. Kranz anatomy is found in -

(1) C_3 plants

(2) C_4 plants

(3) CAM plants

(4) All plants

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109. Calvin cycle occurs in -

(1) C_3 plants only

(2) C_4 plants only

(3) CAM plants only

(4) In all plants

110. Number of ATP and $NADPH.H^+$ required respectively in regeneration step for synthesizing one molecule of sucrose.

(1) 12 ATP, 6 $NADPH.H^+$

(2) 24 ATP, 12 $NADPH.H^+$

(3) 6 ATP, 0 $NADPH.H^+$

(4) 12 ATP, 0 $NADPH.H^+$

111. Which of the following is primary electron acceptor in cyclic photophosphorylation ?

(1) Ferredoxin

(2) Pheophytin

(3) Ferredoxin reducing substance

(4) Plastoquinone

112. RuBisCO show :-

(1) Only carboxylase activity

(2) Only oxygenase activity

(3) Both (1) and (2)

(4) Hydrolase activity

113. Which of the following condition is favourable for photorespiration ?

(1) High light intensity

(2) Low O_2

(3) High CO_2

(4) Low temperature

114. Match the following.

Column-I		Column-II	
(a)	Complex-I	(i)	Succinate dehydrogenase
(b)	Complex-II	(ii)	NADH dehydrogenase
(c)	Complex-IV	(iii)	Cytochrome bc_1
(d)	Complex-III	(iv)	Cytochrome C oxidase

	(a)	(b)	(c)	(d)
(1)	i	ii	iii	iv
(2)	ii	i	iv	iii
(3)	iii	i	ii	iv
(4)	iv	iii	ii	i

115. Which of the following complex contains cytochrome a, a_3 and two copper centres ?

(1) Complex I

(2) Complex II

(3) Complex III

(4) Complex IV

116. **Assertion :** Krebs cycle occurs in mitochondrial matrix.

Reason : All the enzyme of TCA cycle present in membrane of mitochondria.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

117. Which of the following is the site of link reaction ?

(1) Cytoplasm

(2) Mitochondria

(3) Chloroplast

(4) Both (1) & (2)

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118. Which one of the following is not the cofactor of link reaction ?

(1) Mg^{2+} (2) CoA

(3) NAD^{+} (4) Ca^{+2}

119. While the oxygen concentration is low, the product of glycolysis pyruvate is turned into _____.

(1) Acetyl CoA

(2) Ethanol and CO_2

(3) Citric acid

(4) Phosphoenol pyruvate

120. The continued oxidation of acetyl Co-A via the TCA cycle requires the continued replenishment of _____ :

(1) Malic acid

(2) Succinic acid

(3) Oxaloacetic acid

(4) α-ketoglutaric acid

121. Fill in the blank space with the correct option.

_____can be used as respiratory substrates in plants.

(1) Only carbohydrates.

(2) Only proteins and fats.

(3) Only organic acids.

(4) Carbohydrates, proteins, fats and organic acids.

122. The breaking of C-C bonds of complex compounds leading to release of considerable amount of energy is called :-

(1) Photosynthesis

(2) Respiration

(3) Translocation

(4) Transpiration

123. Which of the following statement is not correct for plants.

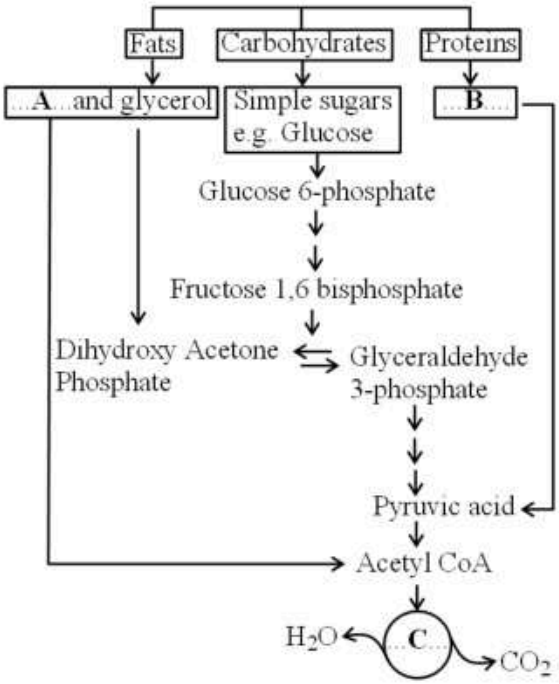
(1) There is very large transport of gases from one plant part to another.

(2) Plants can get along without respiratory organs.

(3) Plants respire at rates for lower than animal do.

(4) the distance that gases must diffuse even in large bulky plants is not great.

124. Identify A, B and C :-



(1) A ⟶ Lipid, B ⟶ Fatty acid, C ⟶ Link reaction

(2) A ⟶ Fatty acids, B ⟶ Amino acids, C ⟶ Kreb's cycle

(3) A ⟶ Monosaccharide, B ⟶ Sugar, C ⟶ Glycolysis

(4) A ⟶ Fatty acid, B ⟶ Lipid, C ⟶ ETS

125. Respiration is a :

(1) Catabolic process.

(2) Anabolic process.

(3) Amphibolic process.

(4) Endergonic process.

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126. **Assertion :-** Plant growth is unique and localised.

Reason :- Plants retain the capacity for unlimited growth throughout their life and that growth occur at specific locations.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

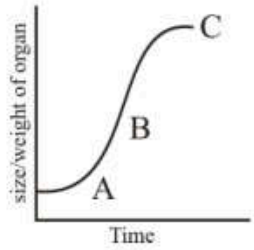
(4) Assertion is False but the Reason is True.

127. Math the following :-

	Match - I		**Match - II**
(i)	Terpenes	(a)	Abscisic acid
(ii)	Adenine derivatives	(b)	Gibberellin
(iii)	Derivatives of carotenoids	(c)	Ethylene
(iv)	Gas	(d)	Cytokinine

	(i)	(ii)	(iii)	(iv)
(1)	a	b	c	d
(2)	b	a	c	d
(3)	b	d	a	c
(4)	c	d	a	b

128. Which of the following is correct regarding this diagram :-



(1) Arithmetic growth
A = Lag phase, B= Exponential phase
C = Stationary phase

(2) Geometric growth
A=Exponential phase, B=Lag phase
C=Stationary phase

(3) Geometric growth
A=lag phase, B=Exponential phase
C=Stationary phase

(4) Relative growth
A=lag phase, B=Exponential phase
C=Stationary phase

129. Which of the following statement is incorrect ?

(1) Cytokinin helps overcome the apical dominance.

(2) Auxin promotes flowering in pineapples

(3) Ethylene also controls xylem differentiation and helps in cell division.

(4) Gibberellin is used to increase the length of grapes stalk.

130. Which of the following hormones induce rooting on stem cuttings ?

(1) Auxin (2) Ethylene

(3) Cytokinin (4) Gibberellins

131. Phytohormone that promotes bolting in many plants with rosette habit is-

(1) Auxin (2) Cytokinins

(3) Gibberellins (4) Abscisic acid

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132. The discovery of gibberellins is associated with a disease of :-

(1) Wheat (2) Rice

(3) Maize (4) Barley

133. Retention of chlorophyll and delayed senescence of leaves is done by

(1) Auxin (2) GA_3

(3) Cytokinins (4) Ethylene

134. __A__Conferred in 1910, the release of a volatile substance __B__ from ripened oranges that hastened the ripening of stored unripened bananas :-

Identify A and B.

(1) (A)-F. skoog, (B)-Auxin

(2) (A)-E. kurosawa, (B)-Gibberellic acid

(3) (A)-H. H. Cousins, (B)-Ethylene

(4) (A)-Miller et. al., (B)-Ethylene

135. Plant hormone that plays an important role in seed development, maturation and dormancy is-

(1) Auxin (2) Ethylene

(3) Cytokinin (4) Abscissic acid

SECTION-B (BOTANY)

136. Find correct statement -

I. In green plants H_2O is hydrogen donor and is oxidised to O_2.

II. Jan Ingenhousz hypothesised that plants restore to the air whatever animal and candle remove.

III. Engelmann provided evidence for production of glucose when plants grow.

(1) I & II

(2) II & III

(3) II only

(4) I only

137. **Assertion** : Except sciophytes and plants of dense forest, light is rarely a limiting factor in nature.

Reason : Light saturation occurs at 10 percent of full summer sun light.

(1) Both assertion and reason are true and the reason is a correct explanation of the assertion

(2) Both assertion and reason are true and the reason is not a correct explanation of the assertion

(3) Assertion is true but reason is false

(4) Both assertion and reason are false

138. What are the products of light reactions ?

(1) ATP and NADPH (2) Sugar

(3) CO_2 (4) All of the above

139. Choose correct option for fill in the blanks-

In half leaf experiment cotton was soaked with __'A'__ and set up was placed in __'B'__.

(1) A–NaCl, B–Light

(2) A–KOH, B–Dark

(3) A–NaOH, B–Dark

(4) A–KOH, B–Light

140. Law of limiting factors was given by :-

(1) Blackman (2) Hill

(3) Calvin (4) Van Niel

141. Fill in the blanks-

_____ also receives reducing equivalents via _____ that is generated during oxidation of succinate.

Select the correct option-

(1) $NADH_2$, Ubiquinone

(2) Ubiquinone, $FADH_2$

(3) Plastoquinone, $FADH_2$

(4) Ubiquinone, $NADH_2$

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142. Enzyme of TCA cycle which is competitively inhibited by malonate.

(1) Malate dehydrogenase

(2) Isocitrate dehydrogenase

(3) Fumarase

(4) Succinate dehydrogenase

143. Yeasts poison themselves to death when the concentration of alcohol reaches about ____ .

(1) 13 %

(2) 7 %

(3) 5 %

(4) 2 %

144. Process in respiration that occurs in anaerobic organisms :

(1) Kreb cycle

(2) Link reaction

(3) Glycolysis

(4) Electron transport system

145. End products of glycolysis are :-

(1) 2 ATP, 2 $FADH_2$ & 2 Pyrvic acid

(2) 2 ATP, 2 NADH & 2 Pyruvic acid

(3) 2 ATP, 2 NADPH & Pyruvic acid

(4) 8 ATP, 2 NADH & 2 Pyruvic acid

146. Which of the following is an assumption of the respiratory balance sheet ?

(1) Respiration is a sequential, orderly pathway.

(2) Only glucose is being respired.

(3) Alternative substrates are entering in the respiratory pathway.

(4) Both 1 and 2

147. Choose the incorrect pair :-

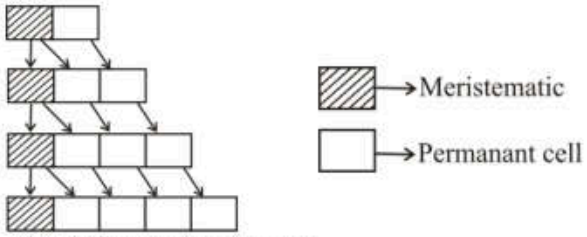
(1) Meristematic phase - Cells are rich in protoplasm posess large conspicuous nuclei.

(2) Elongation phase - Increased vacuolation

(3) Maturation phase - Cell wall are thin and cellulosic

(4) Secondary growth - Lateral meristem.

148.



Above figure represents :-

(1) Zygote division

(2) Arithmetic growth

(3) Geometric growth

(4) Both (1) & (2)

149. Application of which hormone can delay abscission of fruits and leaves at early stage ?

(1) Ethylene

(2) ABA

(3) Auxin

(4) Cytokinin

150. Search for natural substances with cytokinin like activities led to the isolation of _____ from corn-kernels and coconut milk.

(1) Kinetin

(2) Zeatin

(3) Indol-3-acetic acid

(4) Indole butyric acid

SECTION-A (ZOOLOGY)

151. Which structure is tracts of myelinated nerve fibres?

(1) Association areas (2) Hippocampus

(3) Corpus callosum (4) Amygdala

152. Respiratory centre is situated in

(1) Cerebellum (2) Medulla oblongata

(3) Hypothalamus (4) Cerebrum

153. Association area of brain is responsible for all of the following functions except

(1) Intersensory association

(2) Memory

(3) Communication

(4) Emotional and sexual behaviour

154. The correct sequence of meninges of brain from inside to out side in human is:

(1) Duramater – Arachnoid – Piamatar

(2) Arachnoid – Piamatar – duramatar

(3) Piamater – Arachnoid – Duramater

(4) Piamater – Duramater – arachnoid

155. Except one, all are parts of brain stem :

(1) Pons (2) Medulla

(3) Cerebellum (4) Mid brain

156. Amygdala and Hippocampus are the parts of–

(1) Fore brain (2) Mid brain

(3) Hind brain (4) Brain stem

157. The major coordinating centre for sensory and motor signaling is–

(1) Thalamus (2) Pons

(3) Medulla (4) Optic lobes

158. The neural system provides an organized network of __A__ connection for a __B__ coordination. Choose the correct option for A, B to complete the given statement.

(1) A - point to point, B - slow

(2) A - chemical, B - fast

(3) A - point to point, B - fast

(4) A - Chemical, B - slow

159. **Assertion (A) :** Impulse transmission across an electrical synapse is always faster than that across a chemical synapse.

Reason (R) : Transmission of an impulse across chemical synapses is very similar to impulse conduction along a single axon.

(1) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.

(2) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

160. Five events in the transmission of nerve impulse across the synapse are given below

A. Opening of specific ion channels allows the entry of ions, a new action potential is generated in the post-synaptic neuron.

B. Neurotransmitter binds to the receptor on post synaptic membrane

C. Synaptic vesicle fuses with pre-synaptic membrane, neurotransmitter releases into synaptic cleft.

D. Depolarization of pre-synaptic membrane

E. Arrival of action potential at axon terminal.

In which sequence to the events occur ?

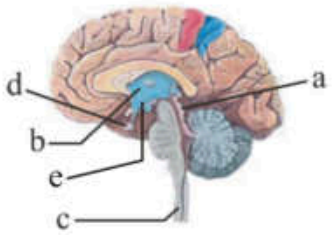
(1) E ⟶ D ⟶ C ⟶ B ⟶ A

(2) A ⟶ B ⟶ C ⟶ D ⟶ E

(3) A ⟶ B ⟶ D ⟶ C ⟶ E

(4) E ⟶ D ⟶ C ⟶ A ⟶ B

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161. 

Choose the incorrect statement from the following

(1) 'a' passes through mid brain

(2) 'c' contains a number of centres which control body temperature

(3) 'b' is a major coordinating centre.

(4) 'a' is a part of the mid brain.

162. Glucocorticoids stimulates :-

(A) Gluconeogenesis

(B) Lipolysis

(C) Proteolysis

(1) A only

(2) A, B and C

(3) B and C only

(4) B only

163. Which of the following is not related to hypersecretion of collip's hormone :-

(1) Osteoporosis

(2) Renal stone

(3) Hypercalcaemia

(4) Tetany

164. Which of the following hormone helps in maintaining the normal rhythms of sleep-wake cycle, body temperature ?

(1) MSH

(2) Melatonin

(3) Thyroxin

(4) (1) & (2) both

165. Adenohypophysis consists of :-

(1) Pars distalis

(2) Pars intermedia

(3) Pars nervosa

(4) (1) & (2) both

166. Deaf mutism & Irregular menstruation can be caused due to :-

(1) Hypersecretion of thyroxine

(2) Hyposecretion of thyroxine

(3) Hyposecretion of sex hormone

(4) Hypersecretion of sex hormone

167. Chemical nature of hormones are :-

(1) Steroid

(2) Amino acid

(3) Proteins

(4) All of the above

168. Which one is not produced by anterior pituitary?

(1) GH

(2) PRL

(3) ACTH

(4) Oxytocin

169. ICSH is another name of–

(1) TSH

(2) FSH

(3) MSH

(4) LH

170. Which of the following hormone does not require secondary messenger for their action?

(1) Gonadotropins

(2) Growth hormone

(3) Epinephrine

(4) Triiodothyronine

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171. Read the following statement and choose the correct statement ?

(1) Hormone are nutrient chemical which act as intercellular messenger.

(2) Each receptor is specific to one hormone only.

(3) Steriod hormone mostly regulate gene expression or chromosome function by the interaction of hormone receptor complex on the membrane.

(4) All hormone are insoluble in water.

172. Read the following column-I & II and identify their correct match :-

	Column-I		Column-II
(A)	Growth hormone	(i)	Pancreas
(B)	Somatostatin	(ii)	Adrenal medulla
(C)	Catecholemines	(iii)	Pineal gland
(D)	Melatonin	(iv)	Pituitary gland
		(v)	Hypothalamus
		(vi)	Adrenal cortex

(1) A-iv, B-v, C-ii, D-iii

(2) A-v, B-iv, C-vi, D-i

(3) A-ii, B-iii, C-iv, D-vi

(4) A-iii, B-i, C-ii, D-iv

173. **Statement-I :** Prolonged hyperglycemia leads to a complex disorder diabetes mellitus.

Statement-II : Glucagon is a hypoglycemic hormone while insulin is a hyperglycemic hormone.

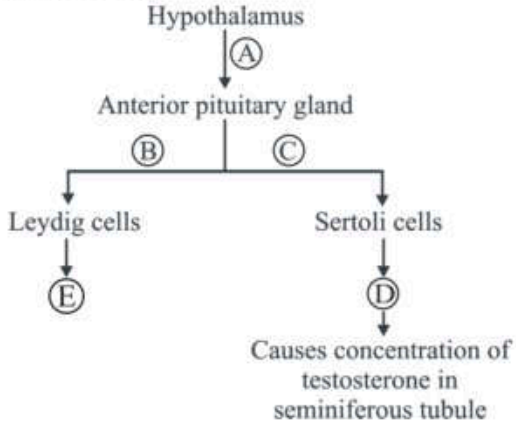
(1) Statement-I and II both are correct.

(2) Statement-I and II both are incorrect.

(3) Statement-I is correct but II is incorrect.

(4) Statement-II is correct but I is incorrect.

174. Observe the following flow chart and choose right statement :-



(1) (B) cause release of (E) and (E) inhibits secretion of (D) (normally)

(2) (C) causes release of (D) and (D) concentrates (E) inside the seminiferous tubule

(3) (A) is GnRH, (C) is ICSH and (B) is FSH

(4) (A) is GnRH, (C) is LH and (B) is FSH

175. Read the following statements :-

(a) Hormones are nutrient chemicals.

(b) Pancreas is organized endocrine gland.

(c) Anterior pituitary is under direct neural regulation of hypothalamus.

(d) Growth hormone is secreted by anterior pituitary.

(e) Deficiency of vasopressin cause acromegaly.

Choose the correct options :-

(1) a,b, & c (2) b,c & d

(3) a,c & d (4) b & d

176. Which of the following terms are "not" related with amoeboid movement ?

(1) Pseudopodia (2) Macrophages

(3) Flagella (4) Neutrophils

177. In the centre of I-Band is an elastic fiber present, which bisect I-band, is called as :

(1) 'H' -zone (2) 'M' -line

(3) 'Z' line (4) 'A' -band

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178. In the resting state a subunit of troponin masks the active binding sites for myosin on :

(1) Actin filaments (2) Meromyosin filament

(3) M–line (4) H–zone

179. A sarcomere consists of

(1) One A-band and two l-band

(2) Half A-band and two half l-band

(3) Half A-band and one l-band

(4) One A-band and two half l-band

180. Match the column I & II with correct option.

	Column-I		Column-II
(A)	Sarcomere	(i)	Red muscle fiber
(B)	Actin filament	(ii)	White muscle fiber
(C)	Slow muscle fiber	(iii)	Contractile unit
(D)	Fast muscle fiber	(iv)	I-band

(1) A-iii, B-iv, C-i, D-ii (2) A-i, B-ii, C-iii, D-iv

(3) A-iv, B-i, C-iii, D-ii (4) A-ii, B-iii, C-iv, D-i

181. Select the correct option :-

(1) First vertebra is the axis and it articulates with the occipital condyles.

(2) Sternum is a flat bone on the dorsal midline of thorax.

(3) Thoracic vertebrae, ribs and sternum together form the rib cage.

(4) 11th and 12th pairs of ribs are called vertebrochondral ribs.

182. At the base of skull, a large foramen is present which allows brain to continue with the spinal cord. That foramen is called :

(1) Foramen ovale (2) Foramen magnum

(3) Vertebral foramen (4) Obturator foramen

183. A single unpaired 'U' shaped bone present in the neck region :

(1) Vomer (2) Hyoid

(3) Mandible (4) Clavicle

184. Which of the following is a synovial joint :

(1) Ball and socket joint (2) Hinge joint

(3) Fibrous joint (4) Both 1 & 2

185. Which of the following not the constituent of axial skeleton :

(1) Skull (2) Patella

(3) Ribs (4) Sternum

SECTION-B (ZOOLOGY)

186. Schwann cell are found in :-

(1) Myelinated nerve fibres of PNS

(2) Unmyelinated nerve fibres of PNS

(3) Both 1 and 2

(4) Myelinated nerve fibers of CNS

187. The components of the human brain responsible for regulating sexual behaviour, excitement, pleasure, rage, and fear are:

(1) Pons & medulla oblongata

(2) Brain stem & Thalamus

(3) Corpus callosum & Motor area

(4) Limbic system & hypothalamus

188. Which of the following structure contains centres, which control respiration, cardiovascular reflexes and gastric secretions:-

(1) Mid brain (2) Cerebellum

(3) Pons (4) Medulla oblongata

189. **Assertion** : Cerebellum has also a very convulated surface.

Reason : Its more surface will accommodate more neurons.

(1) Both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**

(2) Both **Assertion** and **Reason** are correct but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is correct but **Reason** is incorrect.

(4) **Assertion** is incorrect but **Reason** is correct.

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190. Read column-I, II & III and identify their correct match :-

Column-I		Column-II		Column-III	
(A)	Aldosterone	(P)	Alpha cells	(i)	Increases blood glucose levels.
(B)	Epinephrine	(Q)	Beta cells	(ii)	Decreases blood glucose level
(C)	Insulin	(R)	Adrenal medulla	(iii)	Reabsorption of sodium
(D)	Glucagon	(S)	Adrenal cortex	(iv)	Increases heat beat

(1) A-P-i, B-Q-ii, C-R-iii, D-S-iv

(2) A-Q-i, B-P-ii, C-S-iii, D-R-iv

(3) A-Q-i, B-P-iv, C-S-iii, D-R-ii

(4) A-S-iii, B-R-iv, C-Q-ii, D-P-i

191. Progesteron hormone is ?

(1) Stimulates testicular development

(2) Stimulates ovarian development

(3) Stimulates formation of milk alveoli

(4) Stimulates secondary sexual character

192. Nor epinephrine is secreted by :-

(1) Adrenal medulla and increases the heart rate

(2) Adrenal medulla and decreases the heart rate

(3) Adrenal cortex and increases the heart rate

(4) Adrenal cortex and decreases the heart rate

193. In our bodies, what steroid regulates water and electrolytes?

(1) insulin

(2) melatonin

(3) testosterone

(4) aldosterone

194. The secretion of PTH is regulated by :-

(1) Releasing hormone secreted by anterior pitutary

(2) Vit-D_3

(3) Circulating Ca^{++} ion level

(4) All of the above

195. Which of the following hormones controls the functioning of adrenal cortex?

(1) TSH (2) FSH

(3) ACTH (4) LH

196. Select the correct statements :-

(A) Exopthalmic goitre is a form of hypothyroidism

(B) Hypothyroidism may cause menstrual cycle to become irregular.

(C) MSH acts on the melanocytes and regulates pigmentation of the skin.

(D) Adenohypophysis consists of two portions, pairs distalis and pars nervosa.

(1) B, D (2) B, C

(3) A, B (4) A, D

197. Which of the following disorder affects the neuromuscular junction (NMJ) leading to fatigue, weakening and paralysis of skeletal muscle?

(1) Myasthenia gravis

(2) Muscular dystrophy

(3) Osteoporosis

(4) Tetany

198. Which type of joints provide maximum movement to fascilitate the locomotion :

(1) Cartilaginous joint

(2) Fibrous joints

(3) Sutures

(4) Synovial joints

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199. Match List-I with List-II :-

	List-I		List-II
(A)	Scapula	(I)	Acromian
(B)	Floating ribs	(II)	11th and 12th pairs
(C)	Vertebro sternal ribs	(III)	1 to 7th pairs
(D)	Pectoral girdls	(IV)	Scapula, clavicle

Choose the correct answer from the options given below :-

(1) A-III, B-IV, C-II, B-I

(2) A-II, B-IV, C-III, D-I

(3) A-I, B-IV, C-II, D-III

(4) A-I, B-II, C-III, B-IV

200. Match the List I to List II :

List-I		List-II	
A	Myosin	i	Globular protein
B	Actin	ii	Sarcomere
C	Two adjacent	iii	Meromyosin
D	Smooth mussels	iv	Fallopian tube

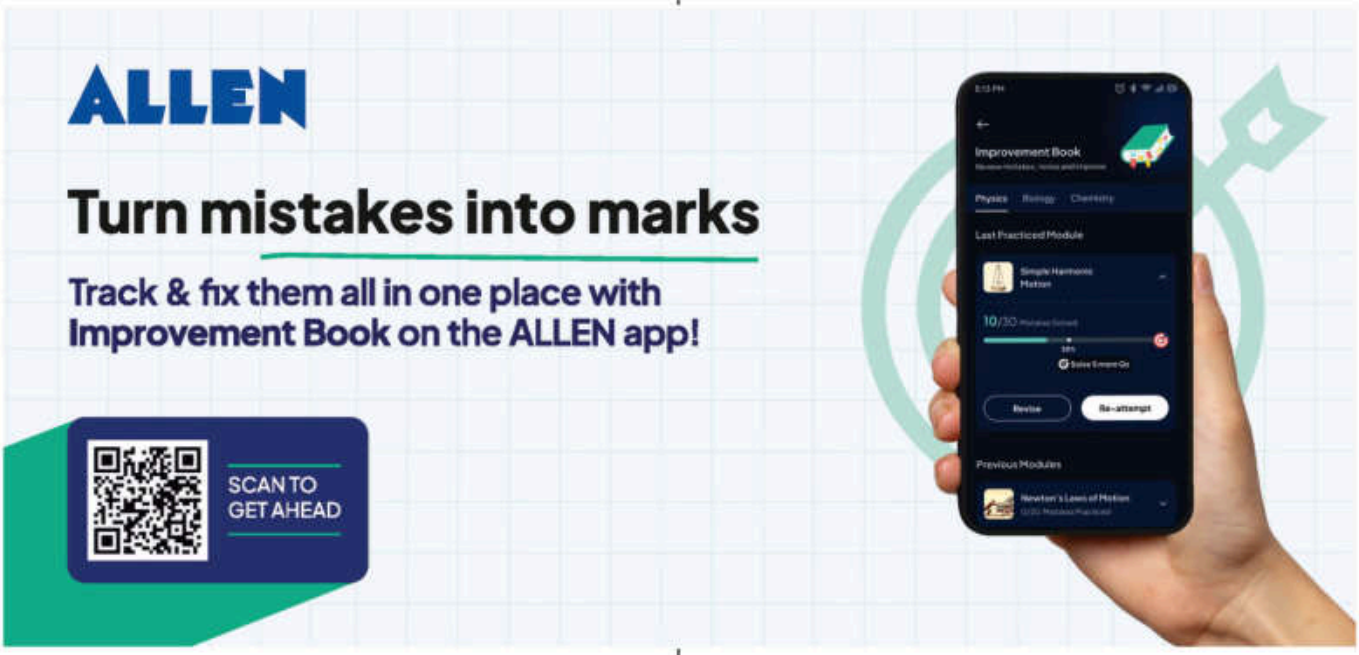
(1) A - iii, B - i, C - ii, D - iv

(2) A - iii, B - ii, C - i, D - iv

(3) A - ii, B - iii, C - i, D - iv

(4) A - iv, B - iii, C - ii, D - i





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